

DISCUSSION

CONCLUDING SESSION

SIR WILLIAM PENNEY, *Chairman*

THE CHAIRMAN: This afternoon, we are going to hear concluding remarks from a number of distinguished speakers from several countries.

I call first on Mr. Moore from the United Kingdom to make his concluding remarks.

R. V. MOORE (United Kingdom): This conference has occurred at a time when five or six countries have formed up their programmes for a final assault, as it were, on fast reactor development. Such is the power of modern technology that success is certain, although no doubt we shall each encounter and have to overcome difficulties on the way. There is a considerable uniformity of view on the approach but there are also important differences, produced mainly by individual national situations.

Sodium cooled fast reactors with cores consisting of subassemblies of fuel pins are the basis of most prototype designs.

Sodium cooled plant certainly has its difficulties, but then there is no such thing anyway as an ideal plant. If engineers of a by-gone generation, designing the first coal-fired boilers, had enjoyed, for example, the benefit of advice from modern corrosion metallurgists, I sometimes wonder whether any such boilers would ever have been built.

The preferred fuel concept at present appears to be uranium or mixed uranium/plutonium oxide, clad in stainless steel and designed so that fission product gases vent to a closed plenum.

The major bar to long burnup appears to be solid fuel swelling, but as experimental data accumulates no doubt designers will be able to find a way round this problem enabling optimum burnup to be approached. The metallurgists, for whose advice we contemporary engineers are eternally grateful, also seem to be finding ways and means of minimizing adverse irradiation effects in fuel cladding and structural materials by heat treatment and by alloying methods.

A considerable amount of new data has been given by the papers in the physics sessions, which will repay careful study and correlation. Compared with a year or so ago, there seems to be greatly increased confidence in the

prediction that large power fast reactors will have substantial negative Doppler coefficients. This is based on experimental work in the United Kingdom and the United States and theoretical work in many countries. A new low power critical facility should become operational in Japan next year, results from which will, no doubt, in due course add to the world's knowledge of fast reactor physics.

The reports from fast reactors already operating were of very great interest. Considering that they were designed as long as ten years ago, it is quite remarkable how well they have performed. In consequence, one has come to accept, with hardly a question, the practicability of pumping 100 tons of sodium round a complex circuit. Only a year or so ago, such problems as maintaining sodium purity at the desired low level were major worries. The kinetic performance of these plants and correlation with the theory of control is very satisfactory. I think, over the next few years, we should see major developments in the important area of reactor instrumentation. It would be reassuring to be able to detect reliably the onset of fuel element failure at an early stage.

Already EBR-II in the United States has demonstrated a delayed neutron monitor system, which has a fairly fast response time. Coupled to charged wire, this could be one method of detecting defective fuel methods. There is some very interesting data from the BR-5 reactor in Russia, covering a three year period of intentional operation with faulty plutonium and uranium oxide fuel elements. They report that sodium logging in the fuel elements did not occur. The information on fission product activities in the coolant and blanket gas is also unique and valuable.

It is an example of the precociousness of the technology that already more advanced fuels are being tested—fuels that have not yet been irradiated in any prototype reactor. Both Russian and British prototype reactors are designed so that a vented-to-coolant concept, in contrast to the present vented-to-plenum, can be tried out.

The next few years no doubt will see the rapid development of sodium components, their simplification and rationalization. Mechanical pumps seem to have displaced electromagnetic pumps for main coolant circulation. In several countries these have satisfactorily completed thousands of hours of testing.

Steam generators are still a worry. We need a better understanding of sodium/water reactions. International exchange of information in this area could be very profitable. Fabrication must be to an exceptionally high standard and the design must enable complete inspection to be certain.

The approach to prototype reactor design in recent years has become more rational and less intuitive. Full-scale design studies of the subsequent industrial plants are carried out before the design of the prototype has frozen. Indeed, these studies are being kept progressively up to date as our understanding

and knowledge improves. There should be much less hesitancy in attaining a fully commercial design early in the exploitation phase than has been the case hitherto.

We in the United Kingdom were also interested to note that several countries, as well as ourselves, are developing what might be called nuclear systems analysis in order to try and establish the economic worth of a given development in advance of its being fulfilled. There are, of course, many pitfalls to this approach. Nevertheless, sensibly used, it is likely to develop into a powerful technique for helping to formulate and discipline applied research programmes. Research and development undoubtedly is very expensive in man-power and resources. Anything which avoids wasted endeavour and helps us to focus on worthwhile advances to be encouraged.

M. SHAW (United States): It is not surprising that the participants in this conference are in mutual agreement and have a high degree of confidence that fast reactors will achieve the dual objective of cheap electricity and conservation of uranium. It is also not surprising that there is mutual agreement on sodium as the priority coolant, on the state of the art, and on the need for planning and executing an extensive research and development programme for mastering this technically difficult and challenging liquid metal cooled fast breeder reactor concept. However, the several different methods being proposed to reach the same objectives do indicate that full agreement has not been reached as to the details and magnitude of the programme needed to meet the objectives in each country.

I believe that we must readily acknowledge such differences, knowing those facets which have made the United States assume our present position, particularly as it relates to the need to strengthen and significantly augment the LMFBR research and development programme before we and our ultimate users, namely, the utility organizations, can be assured that the large number of fast breeder plants built on their systems will be safe, reliable, and economically attractive in a highly competitive commercial environment.

For the sodium cooled fast breeder reactor to compete economically in the electric power business in the United States, it is necessary but not sufficient to restate our objectives for the programme and to develop a plan leading to early construction of a prototype plant. These are the essential steps leading to the more substantive and more critical follow-on actions. We must take positive actions to transfer the ideas and results from research and development programmes into the technical and economic environments of our many competitive designers, fabricators, and utilities; to support and encourage the development of industrial competence which can produce consistently safe, reliable, and economic components, instrumentation, and fuel; to avoid pressures to diversify and splinter our efforts to the degree of compromising adversely our priority programmes; to encourage and help

the utilities to overcome certain apparent prejudices and to accept sodium cooled fast breeder reactors as safe, reliable, and economic additions to their complexes; and to require the engineering organizations responsible for the design and construction of the fast breeder reactor plants to utilize their best talents, resources, and organizational alignments thoroughly to plan and execute the hard commitments which must be made to achieve the programme objectives.

Moreover, an overriding consideration in the entire programme must be the detailed attention to the design and engineering effort with realistic proof tests to verify positively that we have satisfied the technical and engineering prerequisites, and to close that important gap between what we proclaimed could be done and what was actually done.

Further, it will be necessary to demonstrate on an integrated basis and in prototype plants the most successful fuels, components, and systems developed by the research and development programme.

In closing, let me note that we have selected sodium as our priority fast breeder coolant. At this time, it appears unlikely that the economic and technical comparisons will give any significant economic or broad engineering advantages to alternate concepts, such as steam or gas cooled reactors, to the degree necessary to provide reasons for changing the high priority assigned to the sodium cooled concept.

By instituting a thoroughly planned and disciplined effort, taking cognizance of the fundamental factors which contribute to the safety, reliability, and economics, and by continually analysing the causes of troubles which occur, we have the enthusiasm, the determination and the confidence that we can achieve our goal of developing fast breeders which will take their proper place in a timely manner in the future complex of competitive civilian power reactors in the United States and abroad.

A. I. LEIPUNSKII (U.S.S.R.): Fast breeder reactors have become of considerable importance. All countries are interested and, as we have seen at this conference, the exchange of information and of ideas has taken place freely and has been very profitable for the development of our common work.

I would like to make some comments about the questions examined by the conference. For a number of years we have been concerned with the physics of fast breeder reactors and, for that purpose, in addition to a large number of measurements of neutron cross-sections and a study of the processes in atomic nuclei, we have built the BR1 reactor, the BFS assembly which is similar to ZEBRA, and the fast reactor in Dubna. We feel that we are in a position to calculate with sufficient accuracy the critical parameters. The fields in which we have progressed much less are the distribution of the various nuclear processes in reactors and the kinetic operation of reactors which is most important to determine possible modes of failure.

We hope to improve our knowledge with additional information about neutron cross-sections and by accurate determination of the neutron spectra in physical assemblies.

In the near future we hope to start to operate a microtron pulsed source which will enable us to measure the spectra of the assemblies by the "time of flight" method and we hope that this will improve our methods of calculation of reactor parameters.

Another important field which must be developed to a greater extent is the study of the action of irradiation. Although the work accomplished during recent years in the field of long irradiation has brought us many new discoveries, it is such a vast field that it is necessary to develop the work further still, particularly with monocarbides.

A decisive element is the fuel can and it is necessary to understand the effects on the cans under fast reactor conditions. British scientists have obtained remarkable results in this field, but we are not yet quite satisfied because all the results have still to be interpreted in terms of steels and nickel alloys, which must exhibit good performance at fluxes of 10^{16} neutrons/cm²/second, and integral fluxes which reach 10^{24} neutrons/cm². We hope that such materials will be obtained in the near future and that the reliability of the core of the reactor will be sufficiently ensured.

As regards corrosion in sodium, it seems to us that the results already obtained are sufficient in practice, although obviously it may be necessary to study these processes further.

For the reliable operation of a reactor, the important components are the core, consisting of thousands of very fine tubes and the steam generators, the leak tightness of which must be ensured. That is the reason why I would like to stress, especially in the development of fast reactors, the need to obtain tubes of very high quality and to develop inspection procedures applicable to all stages of production. Generally speaking, fast breeder reactors require great technical perfection and we shall have to assist industry in achieving it.

We are constructing a prototype reactor BN350. Work at present has gone smoothly and the building is now 12 metres above ground level. We hope that we shall be able to operate it in 1969. We have great respect for our small reactor, BR5, which has been working for seven years on oxides at a temperature of nearly 500°C and which has provided us with a considerable amount of information. Without giving it up in the future, this year we have started to build an experimental reactor of 60 MW, which will enable us to achieve further development of the core.

A remarkable feature of our conference has been the unanimous agreement reached on the question of strategy for building fast breeder reactors, the choice of fuels, oxides first and the work on carbides and metals later, and the choice of sodium as a coolant.

I would like to stress that in the European projects—and that applies to

our project, and those in France, Euratom, the United Kingdom and Western Germany—great importance is attached to inventory and breeding which will facilitate the beginning of fast reactor development and increase the rate of accumulation of plutonium.

The considerable resemblances between the projects described by the representatives of the different countries show that the technique of fast breeder reactors has already come out of its infancy and the technical decisions have been taken. It is also with pleasure that I note that, in M. de Stordeur's report, we find considerations which, in our opinion, are well founded. We referred to them at the last meeting in Geneva: that is to say, that it is useful and of economic value to start a system of fast breeder reactors with enriched uranium and to change progressively to plutonium. If we are sure of the success of fast breeder reactors—and personally, I have no doubt on the subject—then this will make it unnecessary to develop improved converters.

Fast breeder reactors can in the future meet all the requirements in power for all humanity over a prolonged period, particularly if we take into consideration the enormous reserves of uranium in sea water. Certain scientists have promised to extract this at an economic price. We can assure them they do not need to hurry. With fast breeder reactors available, we can afford to wait.

J. GUERON (Euratom): Out of the four main groups in the world firmly engaged on fast reactor work, the Soviet Union, the United Kingdom, and the United States have working experimental machines. They have studied prototypes and the first two are definitely committed or well engaged in prototype construction. Not yet fully united, Europe is somewhat lagging in achievement, if not in intent or effort.

Next year will see the commissioning of our experimental machines at Cadarache and Karlsruhe. Our intent to aim at early decisions on a prototype is emphasized by many authoritative statements and above all by the project work of which you have heard reports during these 3 days. Let me add that there is so far a very large measure of agreement between the State organizations and industry in Euratom as to the fact that, under our conditions, prototype construction and operation should, earlier rather than later, compel the dovetailing of the many special skills involved in such complex machines. This might then be the last as well as the first conference where such a marked difference will appear between our achievements and those of the other three major groups, but this difference should not obscure the essential concurrence of opinion and design so well expressed this morning by Mr. Carruthers.

Of the two rational ways he proposed to take account of such a situation, namely contraction or concerted or even accentuated diversity, I would like

to favour the latter with its implication of liberal exchange of information as a counter insurance against risks of prototype failure.

It seems clear to me that such an evolution would help to solve two main problems which permanently confront us all. One is that of balance between the scientific and engineering work which is so markedly different inside the various groups. The other is that of giving a fair chance to unorthodox and/or futuristic concepts as well as to renewed attacks on eternal problems such as the permanently unknown water chemistry.

We can all leave with confidence, confirmed in our confidence in the future of fast reactors provided we keep working hard at them.

A. OYAMA (Japan): Dr. Ihara of Japan presented Paper 1/5 in which he pointed out the serious situation with regard to power demand and supply in Japan in the near future. Our incentives for thinking about rather ambitious development programmes on fast breeder reactors are as follows:

First, our serious energy problem which has already been mentioned. Second, we believe that the only way for us to raise our standard of living further is to extend our industrial activity to highly technological fields, like that of the fast breeder. Third, we feel that we should make some contribution to the world-wide efforts on fast breeder development.

In Japan, we now have twenty research or experimental reactors in operation, including seven zero power reactors (following British terminology), and we have 2000 scientists and engineers working for peaceful uses of atomic energy, so that we believe that we can do something to contribute to the power reactor development in the world.

In the policy-making committee for fast breeder reactors, we have been discussing a lot of things since the middle of last year in shaping our own national programme. This is not too easy a task, as probably all of you agree. Senior members of fast breeder clubs in the world began their effort 5–15 years ago. Although we did some fundamental research on fast reactor physics and sodium technology, we did not do any developmental work on fast breeder reactors. What is the most effective way for us? Is there any short-cut way to achieve a large practical fast breeder reactor? In what area of technology shall we concentrate our efforts? There are so many questions in our minds. We feel that there is no short cut to achieving something really useful and we have made up our minds to build up our activity gradually.

In the Japanese Atomic Energy Research Institute, which is our central organization, a 100 MW(th) fast experimental reactor is being designed. The construction of the reactor is expected to be completed in 1971. Its purposes are to provide an irradiation facility of fast reactor fuel and also to gain engineering experience. In the same Institute, we have a fast critical assembly programme and a 2 MW sodium loop programme.

The Atomic Fuel Corporation, the central nuclear fuel organization in

Japan, is now undertaking the fabrication studies of the fuel for the experimental reactor, including plutonium technology. For the confirmation experiment on the fuel we are naturally expecting to have the opportunity to use irradiation facilities in the more advanced countries.

The presentation of the papers and the discussions at the conference gave us some valuable suggestions about our own research and development programmes which have just started. Taking fast reactor physics as an example, this was especially so with regard to sodium void effects and the Doppler effect. In addition, information was given on the heterogeneity effect on the reactivity coefficient. These discussions were very valuable to us in planning our fast critical experiments.

As mentioned by Dr. Hirota in Paper 5A/3, the fast critical assembly (FCA) is now under construction at the JAERI. A major purpose of our critical experimental programme is to predict accurately the neutron spectra, sodium void effect, and Doppler effect in large fast reactors. This facility will go critical next spring with 20% enriched uranium fuel.

During the first stage of experiments, systems to be investigated are small fast systems or fast thermal coupled systems. In the second stage, plutonium systems will be investigated by using the zoned loading.

I realize that our critical experiment programme is, at present, rather far behind. Much more accurate experiments and their correct interpretation, however, are needed in the future, as pointed out in the discussion. In these respects, we believe that there is some room left for us to make a contribution to fast reactor physics.

In conclusion, I hope that the more advanced members of the fast breeder reactor club will give Japan, the youngest member, many suggestions and advice based on their excellent experiences. At the next international conference on this subject, I hope that we can contribute more comprehensive papers and discussions than this time.

Concluding Address by Sir William Penney

We have nearly come to the end of the conference. Over the next few years, when we look backwards in time, we will say that this conference has proved to be an outstanding success. We have heard here that many countries are making notable progress in the development of the fast reactor system. New technological data have been presented on the physics of fast reactors, on irradiation performance of fuel materials and subassemblies, on sodium technology and components, and on engineering developments, including steam generators. These will have a profound influence in shaping the technical policy of future work.

Obviously, from what we have heard, the total cost of fast reactor development around the world is very high, but all of us here—and all of us here are

enthusiasts in nuclear power—believe that the costs are fully justified by sound economic arguments.

It is clear from this conference that the state of knowledge on plutonium/uranium oxide fuel and on sodium technology give the fast reactor based on sodium cooling a clear lead in time over any other type, with gas or steam cooling. Moreover, there is no indication so far that there is sufficient promise in the gas cooled or steam cooled fast reactor to compete economically with the sodium cooled reactor. However sodium cooling has so far commanded most of the effort and the expressions of intent which have been given by speakers from several countries to continue some research and development on gas or steam cooling will clarify these issues over the next few years.

Certainly, the reported results on fuel performance are outstandingly encouraging and in themselves make a substantial advance in the economic promise of the fast reactor system, however cooled.

Fast reactor safety and siting criteria have not formed a primary topic in this conference but we can see that these important topics are being given a lot of attention with promising outcome. Later conferences on fast reactors, wherever they may be held, should give explicit attention to safety and siting. Progress here will be greatly assisted by the operational experience which is now accumulating on experimental fast reactors, and operational experience with fast reactors is another topic which must have a prominent place in future conferences.

Finally, I am sure that everyone present would wish me, as the Chairman of this final session, to thank all those who have helped to make the conference a success. I wish in particular to thank the British Nuclear Energy Society and its Chairman, Dr. Dunworth, for sponsoring the conference, and the Institute of Civil Engineers for making their premises available to us. Particular thanks are due to the Conference Committee and the Conference Secretariat. We are all grateful to the interpreters who have dealt with a difficult task with excellent competence. We thank, too, all those who presented papers or took part in discussions, and those who only listened and learned may be sure that, individually and collectively, they have advanced knowledge and understanding in a happy atmosphere of full and generous international collaboration.

Speaking from the United Kingdom, I would like to thank all the distinguished visitors who have honoured us with their presence here and to thank them for their contributions.